Jannik Birkholz

Electrochemical Engineer

Personal Information

Address
Eduard-Spranger-Straße 11
80935 Munich

Email
jbirkholz21@gmail.com

Phone
+49 1749439937

Date of birth
14.09.1997

LinkedIn
www.linkedin.com/in/jannik-birkholz-6077551aa

Technical Skills & Tools

Measurement Techniques:

Potentiostat (EIS, GCD, CV, LSV), Titration, UV/Vis, RDE

Data/Software:

Python, Excel, R (R-Studio), ChatGPT, Origin

Design:

Prototyping/Test bench setup

Documentation:

SOPs, test protocols, result reports

Through my experience in the development of electrochemical energy storage systems, particularly RedOx flow batteries and hybrid H_2/Fe^{3+} cells, I have successfully accompanied the entire development process from test planning to cell design. In addition to my technical expertise in material and process development, I bring experience in technology and innovation projects as well as strategic responsibility in production decisions and patent applications. With my analytical mindset, hands-on mentality, and interdisciplinary collaboration, I drive innovations that combine performance, reliability, and sustainability.

Professional Experience

04/2022 - present

Working Student and R&D Engineer

Voltstorage, Munich

- Setup and operation of various test stations
- Strategic development decisions
- Development of complementary technologies
- Development of an iron/hydrogen cell
  --> Increased operating current density by >300%
  --> Reduced pressure drop by 50%
  --> Eliminated interval operation; runtime improve from ~60% → 100%

04/2021 - 10/2021

Working Student

VonRoll Automotive, Augsburg

- Electrical and thermal insulation of automotive batteries
- Material and process development
- Sample preparation and prototype construction
- Creation of evaluations and measurement series
- Research of guidelines in the field of electromobility

07/2018 - 12/2019

Internship & Working Student

SGL Carbon, Meitingen

- Material and process development for CFRP reinforcements
- Further development of a forming process for CFRP profiles
- Participation in various development projects
- Prototype construction

Key Projects

10/2023 - 03/2024

Master's Thesis

Voltstorage, Munich

Electrolyte characterization for pump optimization in redox flow batteries

- Objective: Pressure drop/viscosity vs. pump performance; derivation of operating windows
- Contribution: Experimental design, data acquisition (viscosity/conductivity), pulsed charge/discharge, EIS, evaluation & recommendations

03/2023 - 08/2023

Research Internship

National Taiwan University, Taipei

MOF coating for NMC cathodes in Lithium-Ion-Batteries

- Objective: Increased stability/SEI control; coating processes & electrochemical tests
- Contribution: Testing and evaluation of various coating methods

Languages

German - Native

English - C2

Certificates & Engagement

2nd Bundesliga Volleyball

Sport and Leadership coaching

HAW.International Scholarship

10/2022
- 03/2023

Research Project

TUM, München

Membrane dry-out effects in an electrolysis cell
Objective: Influence of flow field design on humidification, cell resistance, and stability
Contribution: Setup/experiments, data analysis

10/2020
- 03/2021

Bachelor's Thesis

Frauenhofer IGCV, Augsburg

Process optimization of an SLM metal 3D printer
Objective: Increase printing speed without defects/air inclusions
Contribution: Material and process analysis, printing speed increased by 60% while maintaining quality

Education

09/2021
- 03/2024

M.Sc. Chemical Engineering

TUM, Munich

Semester abroad: Research internship in Taipei

10/2016
- 04/2022

B.Eng. Energy and Process Engineering

University of applied Sciences, Augsburg

Semester abroad: Study semester in Chicago